Model Validation Report

Retail Unsecured Loan PD Scorecard

Item	Detail
Model ID / version	CR-PD-001 / 2.3
Model type	Credit risk — probability of default (PD)
Model owner	Retail Credit Risk Modeling
Validation run	20261023-194444-CR-PD-001-c39695
Automated analysis	2026-10-23 19:44 UTC (4.9 s)
Reviewed and signed off by	Demo reviewer, 2026-10-23 19:44 UTC
Overall rating	Unsatisfactory
Framework	Federal Reserve / OCC SR 11-7 — Supervisory Guidance on Model Risk Management

1. Executive summary

Overall rating: Unsatisfactory

The model should not be relied on for its intended use until High-severity findings are remediated, or only under compensating controls approved by the model risk committee.

The validation of Retail Unsecured Loan PD Scorecard concludes with an overall rating of Unsatisfactory. The model should not be relied on for its intended use until High-severity findings are remediated, or only under compensating controls approved by the model risk committee. After review, 14 finding(s) were retained: 5 High, 8 Medium and 1 Low severity. The most material are Multicollinearity among model features (severe) (AS-001); Model under-predicts defaults (BT-001); Values outside documented valid ranges in development data (DQ-002).

Executive summary drafted by template from the reviewed findings; ratings and severities are set by the validator.

Severity	Findings	Remediation due within
High	5	90 days
Medium	8	180 days
Low	1	365 days

Review status: 13 accepted, 1 edited, 1 rejected, 0 pending.

2. Model and validation scope

Purpose: Estimate 12-month probability of default for unsecured personal loan applicants.

Intended use: Application approval cut-offs and risk-based pricing; input to CECL PD estimates.

Documented assumptions tested:

- Log-odds of default are linear in the (transformed) features.
- Development sample is representative of the current applicant population.
- Features are available at application time.
- Default definition: 90+ days past due within 12 months of origination.

Documented limitations:

- Not validated for small-business or secured lending.

Validation activities

Review area	SR 11-7 element	Status	Findings proposed
Data Quality	Evaluation of conceptual soundness, Ongoing monitoring	Completed	8
Assumptions	Evaluation of conceptual soundness	Completed	3
Backtesting	Outcomes analysis	Completed	2
Challenger Benchmarking	Ongoing monitoring	Completed	2
Sensitivity Analysis	Evaluation of conceptual soundness	Completed	0

Package validated (SHA-256 of each file, first 12 characters): model.json cfaa3e448304; model_package.yaml 00e5f78a0a77; test.csv b094907544c0; train.csv 02df3812dc98

3. Findings summary

ID	Severity	Finding	Area	Review	Due
AS-001	High	Multicollinearity among model features (severe)	Assumptions	Accepted	2027-01-21
BT-001	High	Model under-predicts defaults	Backtesting	Accepted	2027-01-21
DQ-002	High	Values outside documented valid ranges in development data	Data Quality	Accepted	2027-01-21
DQ-003	High	Suspected target leakage	Data Quality	Accepted	2027-01-21
DQ-004	High	Feature distribution shift between development and validation samples (significant)	Data Quality	Accepted	2027-01-21
AS-002	Medium	Non-linear relationship with log-odds	Assumptions	Accepted	2027-04-21
BT-002	Medium	Predicted PDs do not match observed default rates across risk deciles	Backtesting	Accepted	2027-04-21
CH-001	Medium	Gradient boosting challenger outperforms the champion	Challenger Benchmarking	Accepted	2027-04-21
CH-002	Medium	Random forest challenger outperforms the champion	Challenger Benchmarking	Accepted	2027-04-21
DQ-005	Medium	Missing values in development data	Data Quality	Accepted	2027-04-21
DQ-006	Medium	Duplicate records in development data	Data Quality	Accepted	2027-04-21
DQ-007	Medium	Extreme outliers in development data	Data Quality	Accepted	2027-04-21
DQ-008	Medium	Default rate differs materially between samples	Data Quality	Accepted	2027-04-21
AS-003	Low	Features without statistically significant effect	Assumptions	Edited	2027-10-23

4. Detailed findings

Data Quality

The Data Quality review of Retail Unsecured Loan PD Scorecard identified 7 finding(s) (3 high, 4 medium severity). High-severity items: Values outside documented valid ranges in development data; Suspected target leakage; Feature distribution shift between development and validation samples (significant).

DQ-002 · Values outside documented valid ranges in development data

Severity: High · Evaluation of conceptual soundness · Review: accepted · Remediation due 2027-01-21

Impossible or undocumented values found: age (12 rows outside [18.0, 100.0]).

Evidence

- dataset: development
- violations: {age: {count: 12, allowed: [18, 100], observed_min: -1, observed_max: 150}}

Recommendation: Trace the records to source, correct or exclude them, and add range validation to the data pipeline.

DQ-003 · Suspected target leakage

Severity: High · Evaluation of conceptual soundness · Review: accepted · Remediation due 2027-01-21

On its own, recovery_amount (AUC 1.000) separates defaulters from non-defaulters almost perfectly. A feature this predictive is usually only observable after the outcome.

Evidence

- univariate_auc: {recovery_amount: 1}
- threshold: 0.95

Recommendation: Confirm each feature's availability at the decision (application) date; remove any post-outcome variable and re-estimate the model.

DQ-004 · Feature distribution shift between development and validation samples (significant)

Severity: High · Ongoing monitoring · Review: accepted · Remediation due 2027-01-21

Population Stability Index above 0.25: credit_utilization (PSI 0.662), utilization_6m_avg (PSI 0.449).

Evidence

- psi: {credit_utilization: 0.6625, utilization_6m_avg: 0.449}
- threshold: 0.25

Recommendation: Investigate drivers of the shift (policy, channel, macro) and confirm the model remains fit for the current population; re-test performance on the shifted segment.

DQ-005 · Missing values in development data

Severity: Medium · Evaluation of conceptual soundness · Review: accepted · Remediation due 2027-04-21

1 column(s) in the development sample contain missing values: annual_income (7.9%).

Evidence

- dataset: development
- missing_rate: {annual_income: 0.0788}

Recommendation: Confirm the root cause of missingness (system outage, optional field, data feed), document the treatment applied by the developer, and test whether missingness is correlated with the target or model output.

DQ-006 · Duplicate records in development data

Severity: Medium · Evaluation of conceptual soundness · Review: accepted · Remediation due 2027-04-21

139 of 8160 rows (1.70%) in the development sample are exact duplicates.

Evidence

- dataset: development
- duplicate_rows: 139
- rows: 8160
- rate: 0.017

Recommendation: Remove duplicates before estimation and re-run development tests; duplicated records overweight some observations and understate standard errors.

DQ-007 · Extreme outliers in development data

Severity: Medium · Evaluation of conceptual soundness · Review: accepted · Remediation due 2027-04-21

Values more than 6 robust standard deviations from the median: annual_income (120 rows).

Evidence

- dataset: development
- z_threshold: 6
- features: {annual_income: {count: 120, share: 0.0147, max_value: 1e+08}}

Recommendation: Determine whether these are data-entry errors or genuine extremes; document the capping or winsorisation rule and test model sensitivity to it.

DQ-008 · Default rate differs materially between samples

Severity: Medium · Ongoing monitoring · Review: accepted · Remediation due 2027-04-21

Default rate moved from 12.71% (development) to 21.03% (validation).

Evidence

- train_rate: 0.1271
- test_rate: 0.2103

Recommendation: Assess whether the shift reflects the credit cycle or a population change; consider recalibration.

Assumptions

The Assumptions review of Retail Unsecured Loan PD Scorecard identified 3 finding(s) (1 high, 1 medium, 1 low severity). High-severity items: Multicollinearity among model features (severe).

AS-001 · Multicollinearity among model features (severe)

Severity: High · Evaluation of conceptual soundness · Review: accepted · Remediation due 2027-01-21

Variance inflation factors above 10: credit_utilization (VIF 35.6), utilization_6m_avg (VIF 35.6). Coefficients of collinear features are unstable and their signs cannot be interpreted.

Evidence

- vif: {credit_utilization: 35.6, utilization_6m_avg: 35.6}
- threshold: 10
- correlations: {credit_utilization ~ utilization_6m_avg: 0.986}

Recommendation: Keep one feature from each collinear group (or combine them), re-estimate, and confirm the remaining coefficients have the expected signs.

AS-002 · Non-linear relationship with log-odds

Severity: Medium · Evaluation of conceptual soundness · Review: accepted · Remediation due 2027-04-21

Adding a squared term significantly improves fit (Bonferroni-adjusted) for: annual_income (adjusted p < 0.0001), debt_to_income (adjusted p < 0.0001). This contradicts the documented assumption that log-odds are linear in the features.

Evidence

- lr_test_p_values: {annual_income: 0, debt_to_income: 0}

Recommendation: Consider binning (WoE), a transformation or a spline for these features.

AS-003 · Features without statistically significant effect

Severity: Low · Evaluation of conceptual soundness · Review: edited · Remediation due 2027-10-23

In the independent re-estimation these features are not significant at 5% (|z| < 1.96): credit_utilization (z = 0.97), num_open_accounts (z = 0.14), loan_amount (z = -0.26).

Evidence

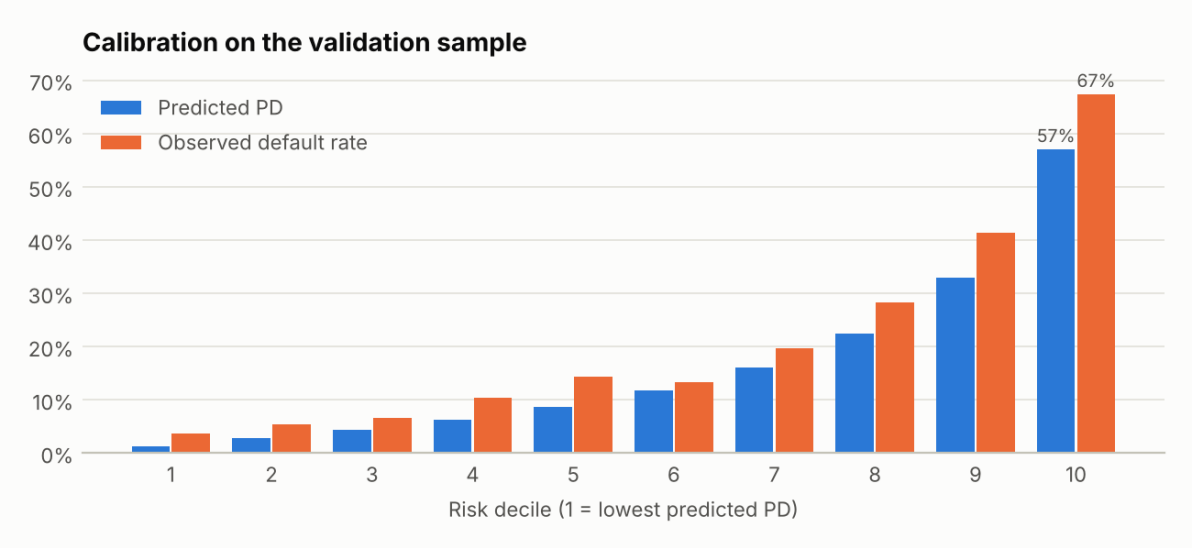
- z_statistics: {credit_utilization: 0.97, num_open_accounts: 0.14, loan_amount: -0.26}

Recommendation: Resolve together with AS-001 (multicollinearity).

Validator note (Demo reviewer): Insignificance of utilisation is a symptom of the collinearity in AS-001.

Backtesting

The Backtesting review of Retail Unsecured Loan PD Scorecard identified 2 finding(s) (1 high, 1 medium severity). High-severity items: Model under-predicts defaults.



Mean predicted PD and observed default rate by decile of predicted PD.

BT-001 · Model under-predicts defaults

Severity: High · Outcomes analysis · Review: accepted · Remediation due 2027-01-21

631 defaults observed against 491 predicted (observed rate 21.03% vs mean PD 16.35%; z = 7.81, p = 5.7e-15). Under-prediction is non-conservative: losses, pricing and reserves would be understated.

Evidence

- observed_defaults: 631
- expected_defaults: 490.6
- calibration_z: 7.81
- calibration_p: 0

Recommendation: Recalibrate the intercept to the current default rate (or a through-the-cycle target) and document the calibration target.

BT-002 · Predicted PDs do not match observed default rates across risk deciles

Severity: Medium · Outcomes analysis · Review: accepted · Remediation due 2027-04-21

Hosmer-Lemeshow test on the validation sample: statistic 78.6, p = 9.3e-14.

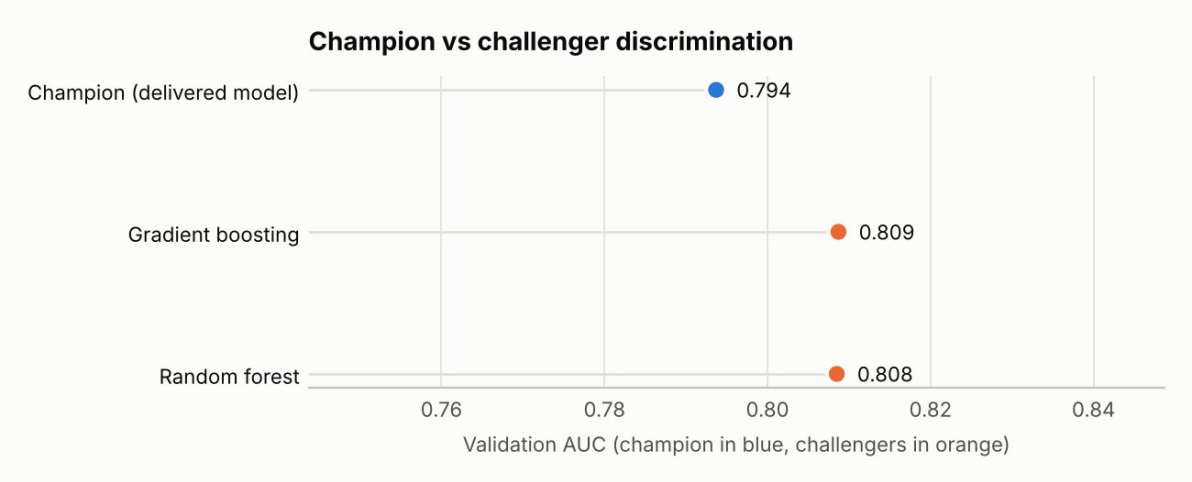
Evidence

- hosmer_lemeshow: {stat: 78.61, p: 0}
- deciles: [{decile: 1, n: 300, predicted_rate: 0.0121, observed_rate: 0.0367}, {decile: 2, n: 300, predicted_rate: 0.0281, observed_rate: 0.0533}, {decile: 3, n: 300, predicted_rate: 0.0441, observed_rate: 0.0667}, …]

Recommendation: Review the decile table for where the model mis-states risk; recalibrate by segment if needed.

Challenger Benchmarking

The Challenger Benchmarking review of Retail Unsecured Loan PD Scorecard identified 2 finding(s) (2 medium severity).



Validation-sample AUC of the delivered model and the validator's challengers.

CH-001 · Gradient boosting challenger outperforms the champion

Severity: Medium · Ongoing monitoring · Review: accepted · Remediation due 2027-04-21

On the validation sample the gradient boosting challenger reaches AUC 0.809 vs 0.794 for the delivered model (gap +0.015, paired-bootstrap 95% interval +0.004 to +0.025). The challenger uses the same features, so the champion's functional form is leaving predictive information unused.

Evidence

- auc: 0.8087
- brier: 0.1259
- auc_gap: 0.015
- auc_gap_95ci: [0.0042, 0.0251]
- rank_correlation: 0.867

Recommendation: Identify where the challenger gains (e.g. partial dependence of the main features) and capture it in the champion through binning or transformations, or document why the simpler model is preferred despite the gap.

CH-002 · Random forest challenger outperforms the champion

Severity: Medium · Ongoing monitoring · Review: accepted · Remediation due 2027-04-21

On the validation sample the random forest challenger reaches AUC 0.809 vs 0.794 for the delivered model (gap +0.015, paired-bootstrap 95% interval +0.003 to +0.025). The challenger uses the same features, so the champion's functional form is leaving predictive information unused.

Evidence

- auc: 0.8085
- brier: 0.1288
- auc_gap: 0.0148
- auc_gap_95ci: [0.0031, 0.0253]
- rank_correlation: 0.835

Recommendation: Identify where the challenger gains (e.g. partial dependence of the main features) and capture it in the champion through binning or transformations, or document why the simpler model is preferred despite the gap.

Sensitivity Analysis

The Sensitivity Analysis review of Retail Unsecured Loan PD Scorecard identified no exceptions.

Appendix A. Findings rejected by the validator

ID	Proposed severity	Finding	Reviewer	Rationale
DQ-001	High	Conflicting records share an ID in development data	Demo reviewer	Conflicting IDs arise from the duplicate rows already raised in DQ-006; tracked there.

Appendix B. Audit trail and turnaround

Measure	Value
Automated analysis	4.91 s
Data Quality	0.176 s
Assumptions	0.064 s
Backtesting	0.105 s
Challenger Benchmarking	4.477 s
Sensitivity Analysis	0.08 s
Review decisions recorded	15
Active review (first to last decision)	0.0 min
Run completion to sign-off	0.0 h
Total elapsed (start to sign-off)	0.0 h

Times are taken from the run's append-only audit log (audit.jsonl), which also records every review decision with reviewer, timestamp and rationale.

Sign-off comment: Illustrative review of a synthetic model package.

Appendix C. Limitations of this validation

- Automated tests cover the checks listed in section 2; they do not replace the validator's judgement on conceptual soundness, business use or the appropriateness of the model's design.
- Statistical tests reject a correct model at roughly their significance level; findings are therefore reviewed individually before they are accepted.
- AI-written text (area summaries, commentary, executive summary) is generated from the computed results only and is labelled as such; it cannot add, remove or re-grade findings.